E18.5 mouse DRG dissociation for FACS, 060214

Masato Hoshi, MD, PhD,
Sanjay Jain Lab, Renal Division, Department of Medicine,
Washington University School of Medicine

Reagents;

- Hibernate-E; Life Technologies, #A12476-01
- DMEM/F12; Life Technologies, #11330
- 0.05% Trypsin/0.02% EDTA; Sigma, #59417C or #T3924
- DNaseI; Sigma, #D5025
- Collagenase; Sigma, #C0130
- RNase inhibitor; RNaseOUT (Life Technologies, 40units/ul) or equal
- 7-AAD; Life Technologies, #A1310 (powder, dissolve in DMSO at 1mg/ml)

Procedures;

1) Euthanize a pregnant mouse following your animal protocol approval and extract embryos from the uterus.
2) Save the embryos/fetuses in a 12-well plate with 3ml of Hibernate-E medium (Life Technologies) in each well of the plate on ice.
3) Take one embryo in 10cm plastic plate with dissection medium (DMEM/F12 (Life Technologies) or Hibernate-E (30ml)) and extirpate DRGs from embryos. (Th11 to S1, total 20 DRGs from each embryo; or as needed for your study)
   (*when you use DMEM/F12, the medium should be put in a CO_2 incubator (cell culture, 5% CO2 incubator at 37C is fine) for more than 2 hours to keep the pH balanced (gauge by the color, prefer orange).
4) Collect DRGs in a 1.5ml tube on ice with dissection medium in a pipette tip. Keep the tube on ice until all dissections are done.
5) Centrifuge at 500 xg for 5min at 4C and discard as much supernatant as possible by using a pipette.
6) Add 500ul (to 1ml, depending on the number of DRGs) of 0.05% Trypsin/ 0.02% EDTA (Sigma) with 200ug/ml of DNaseI (Sigma) to the tube and briefly vortex to mix.
7) Incubate it at 37C for 15min.
8) Triturate with P-1000, then P-200 pipette and dissociate the specimens. Still see some chunks.
9) Add 500ul of of DMEM/F12 + 10% FCS (filtered) and mix well by pipetting.
10) Centrifuge at 500 xg for 5min at 4C and discard supernatant by using a pipette.
11) Add 400ul of DMEM/F12 + 10% FCS (filtered), mix well by pipetting, and then add 600ul of collagenase solution (Sigma, dissolved in DMEM/F12 + 10%FCS (sterilized) at 2mg/ml; final collagenase concentration is 1.2mg/ml). Mix well by pipetting.
12) Incubate it at 37C for 10min.
13) Triturate with P-200 pipette till you notice complete dissociation of the specimen.
14) Centrifuge at 500 xg for 5min at 4C and discard supernatant by using a pipette. Leave around 30ul to avoid disturbing the cell pellet.

15) Add 500ul of PBS + 5% FCS (filtered) with 0.1units/ul of RNase inhibitor (Life Technologies, 40units/ul, 25ul to 10ml PBS/FCS solution) and mix well by pipetting.
16) Centrifuge at 500 xg for 5min at 4C and discard supernatant by using a pipette.
17) Repeat this washing step twice more with PBS/FCS/RNase inhibitor solution. Total three times. In the last wash, the cells get dispersed very quickly with only few rounds of pipetting.
18) After discarding as much supernatant as possible, add 300ul (or appropriate volume for FACS) of PBS + 5% FCS (filtered) with 0.1units/ul of RNase inhibitor and mix well by pipetting.
19) Filter the dissociated cells with 40um pore size cell strainer (BD, #352340) and collect the cells into a FACS tube (BD, #352063).
20) Add 7-AAD at 1ug/ml concentration, if needed, and do sorting. 7-AAD is added 10min before sorting. We have had good success using MoFlo, Sony or AriaII, at 100 um nozzle size, and 30psi.
21) Sorted cells are collected in 750ul of TRIzol-LS (Life Technologies, #10296-010) for RNA extraction and saved in -80C until use.